

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

SKILLS

- **Machine Learning / Statistics:** Classification, Regression, Feature Engineering, Model Evaluation, Precision, Recall, Accuracy, Statistical Analysis, Predictive Analytics, Data Validation
- **Programming / Querying:** Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Neo4j, Scikit-learn, LightGBM
- **Visualization / Reporting:** Power BI, Tableau, DAX, Excel, VBA, Dashboard Development, Data Visualization
- **Data / Cloud / Tools:** AWS S3, EC2, Redshift, EKS, FastAPI, Docker, Git, Kafka, pgVector, GCP

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

- Designed and implemented end-to-end machine learning pipeline to analyze multi-dimensional behavioral data from 100+ users in a healthcare context, applying feature engineering, statistical modeling, and model-ready transformations to surface actionable insights for downstream product and research stakeholders
- Developed and optimized **PostgreSQL** data models, indexing strategies, and query execution plans for high-volume datasets, improving query performance by **38%** while enabling efficient feature extraction for machine learning workflows
- Developed and implemented state-of-the-art AI retrieval and summarization model using embedding techniques, improving response accuracy by 43% and enabling scalable, context-aware analysis of complex multi-dimensional datasets

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

- Developed, documented, and deployed NLP-based machine learning models in Python to classify and analyze large-scale text datasets in healthcare research contexts, applying advanced statistical preprocessing and feature extraction to reduce processing latency by 41% and improve classification accuracy
- Built distributed **data pipelines** using **Kafka** and **Neo4j** Streams to enable **scalable ingestion**, transformation, and real-time analysis of high-volume unstructured datasets
- Designed production-grade inference **APIs** and deployment workflows to support reproducibility, monitoring, and consistent evaluation of machine learning outputs

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

- Built and deployed production Retrieval-Augmented Generation (RAG) system using state-of-the-art embedding techniques and vector search, combining data pipeline design, feature engineering, and retrieval optimization, then presented findings and system performance to stakeholders to inform product development decisions, improving retrieval efficiency by 64%
- Maintained technical documentation for data models, ML pipeline architecture, and algorithm logic, establishing clear standards for reproducibility and enabling cross-functional engineering and product teams to build on existing work
- Optimized indexing strategies and data processing workflows to enable low-latency querying and scalable deployment of AI-driven systems

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

- Analyzed large-scale operational datasets using **Python** and **PySpark** to identify trends, anomalies, and performance issues, combining exploratory analysis and root cause investigation to support data-driven decision-making
- Built **Power BI** dashboards to track key performance indicators in operational performance, reducing reporting time by 40%

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

- Developed and maintained data models and visualizations for enterprise clients across Sales, Finance, and Operations using **SQL**, **DAX**, and **Tableau**, translating complex multi-dimensional datasets into clear, actionable insights and presenting findings to stakeholders across organizational levels
- Performed data validation, cleaning, and transformation on large multi-source datasets using **SQL** and **Excel**, improving data accuracy by over 30% and reducing reporting inconsistencies, enabling faster and more reliable decision-making across sales, finance, and operations teams

Stu/Dio at Illinois

August 2025 - Present

Project Manager

- Led a **cross-functional** team of six to deliver and scale Deepcover game enhancements by leveraging user data, performance metrics, and feedback analysis to drive feature prioritization and improve games outcomes

PROJECTS

Scalable ML System for Customer Risk Prediction

- Built **LightGBM**-based machine learning model to **predict customer credit risk** from **time-series transaction data**, combining feature engineering, classification modeling, and evaluation techniques to identify high-risk and low-risk segments
- Evaluated model performance using precision, recall, and accuracy metrics, designed a decision framework translating predictions into actionable credit limit recommendations, and produced clear documentation and stakeholder-ready reports, findings and model logic